

A Quantitative Gap Audit of LLM-era NLP for Education: Validity, Deployment, and Multilingual Equity

Abstract

Despite rapid advances in large language model (LLM) capabilities, it remains unclear whether progress in educational NLP translates into measurable improvements in student learning. We present a systematic analysis of 137 papers (2022–2025) across four sectors—assessment and grammatical error correction, feedback and writing support, content and question generation, and tutoring and dialogue systems—introducing three formal gap metrics: the Validity Gap (VG), measuring the proportion of studies that omit learning-outcome evaluation; the Deployment Readiness Index (DRI), capturing real-world evaluation extent; and the Multilingual Equity Index (MEI), quantifying language coverage. We find $VG = 0.927$, $DRI = 0.126$, and $MEI = 0.126$ field-wide. Within grammatical error correction, no study demonstrates learning improvement ($VG = 1.000$). DRI correlates positively with learning-outcome evaluation ($\rho = 0.425$, $p < 0.001$), indicating that real-world deployment and educational accountability are structurally linked. Bengali, Hindi, Urdu, Tamil, and Nepali—collectively serving over 600 million school-age learners—have no dedicated educational NLP system in our corpus ($MEI = 0.0$). Our taxonomy, metrics, and five-gap research agenda constitute a reproducible audit framework for educational NLP accountability.

1 Introduction

Education is a high-impact NLP domain where systems such as grammatical error correction and tutoring dialogue can directly influence student learning. Recent years have seen a rapid shift toward large language models (LLMs) driven by instruction tuning (Ouyang et al., 2022) and advanced reasoning (OpenAI, 2023), alongside open-weight models enabling broader access (Touvron et al., 2023; Jiang et al., 2023).

Progress in capability has not been matched by evaluation. Educational NLP relies heavily on proxy metrics such as $F_{0.5}$ (Bryant et al., 2019) that do not measure actual learning (Ferris, 2004; Truscott, 1996). Learning science highlights deeper cognitive processes ignored in current evaluations (Kapur, 2016), and the field remains heavily English-centric with limited South Asian coverage (UNESCO, 2023).

This paper makes five contributions: (1) a systematic taxonomy of 137 papers (2022–2025) across four sectors and 14 sub-categories ($\kappa \geq 0.79$); (2) three formal metrics VG, DRI, MEI applied across all sectors; (3) a temporal analysis of the LLM transition; (4) the first quantitative analysis of South Asian underrepresentation; and (5) a five-gap research agenda for future work.

2 Related Work

Prior educational NLP work falls into three categories, none of which introduce formal gap metrics or analyse the post-2022 transition.

Pre-LLM task surveys. Surveys on automated essay scoring (Yannakoudakis et al., 2011; Li and Ng, 2024a), GEC (Bryant et al., 2023), and question generation (Heilman and Smith, 2010; Pan et al., 2019) identify evaluation inconsistency but predate instruction-tuned LLMs and omit deployment or learning-outcome analysis.

Broad educational NLP reviews. General surveys and BEA overview papers are primarily descriptive and lack formal frameworks for deployment, learning impact, or equity.

LLM-era analyses. Dong et al. (2024); Min et al. (2022) provide LLM evaluation insights; Kasneci et al. (2023) discuss educational implications. None conduct systematic corpus-level analysis or introduce reproducible metrics. Our work differs by (1) targeting 2022–2025, (2) introducing

VG, DRI, and MEI computed across a controlled corpus, and (3) quantifying South Asian multilingual inequity as a measurable phenomenon.

3 Methodology

3.1 Search Protocol

We conducted a systematic search across four primary sources: the ACL Anthology (ACL, EMNLP, NAACL, EACL, COLING, and co-located workshops including BEA 2022–2025), the Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2022–2025), and arXiv (cs.CL, cs.AI) for preprints with ≥ 15 citations as of March 2025, providing a community-recognition threshold consistent with prior NLP survey practice¹

Search terms combined educational domain keywords (*education, learning, student, assessment, tutoring, grammatical error correction, readability, question generation, essay scoring, feedback*) with methodological terms (*NLP, LLM, large language model, transformer*). Papers were included if they: (1) proposed or evaluated a computational system addressing an educational language task; (2) were published between January 2022 and March 2025; and (3) appeared in a peer-reviewed venue or met the arXiv citation threshold above.

Screening flowchart. The initial query returned 312 candidate papers. After deduplication across sources (43 removed), title-and-abstract screening (89 removed), and full-text review against the exclusion criteria below (43 removed), we obtained a final corpus of **137 papers**.

3.2 Inclusion and Exclusion Criteria

We excluded: (i) papers addressing educational administration without an NLP contribution; (ii) pure dataset papers without modelling; and (iii) papers where the educational framing was incidental to a broader NLP contribution (e.g., a general reading-comprehension benchmark that happened to use school-level passages).

3.3 Annotation Scheme

Each paper was annotated on five primary dimensions:

¹We required 15 citations as a community-recognition proxy, ensuring preprints had received sufficient peer attention to be considered established contributions. Piloting on 20 borderline cases confirmed all papers above this threshold were independently judged relevant by both annotators.

1. **Sector** four categories (see Section 4);
2. **Model type** LLM / BERT-era / Rule-based / Other;
3. **Learning outcome** YES: empirical learning measurement (pre/post-test, controlled experiment, or longitudinal study); NO: proxy metrics only;
4. **Fairness** YES: explicit fairness/equity analysis; NO: absent;
5. **Deployment type** Offline / Simulated / User study / Production.

Inter-annotator agreement. A second annotator labelled a stratified sample of 30 papers (21.9%) on all five dimensions. Cohen’s κ : sector $\kappa = 0.95$; learning outcome $\kappa = 1.00$. Fairness initially yielded $\kappa = 0.56$; after tightening the criterion to require an *explicit* fairness analysis, post-resolution agreement on 20 held-out papers reached $\kappa = 0.81$ (substantial). Remaining disagreements were resolved by discussion.

3.4 Three-Metric Gap Framework

Validity Gap (VG).

$$VG(s) = 1 - \frac{L(s)}{N(s)} \quad (1)$$

where $N(s)$ is the number of papers in sector s and $L(s)$ is the count of papers annotated YES on the learning-outcome dimension (dimension 3, Section 3.3), i.e., those reporting an empirical pre/post-test, controlled experiment, or longitudinal learning measurement.

Deployment Readiness Index (DRI). We assign each paper a continuous deployment score: offline benchmark = 0.00; crowdworker or LLM-simulation = 0.25; small user study ($n < 20$) = 0.50; large user study ($n \geq 20$) = 0.75; production system = 1.00.

$$DRI(s) = \frac{1}{N(s)} \sum_{p \in s} d(p) \quad (2)$$

where $d(p)$ is the deployment score of paper p . Boundaries follow clinical-trial phase criteria (Friedman et al., 2010); shifts of ± 0.10 changed no paper’s tier.

Multilingual Equity Index (MEI). We assign each paper a language-equity score: English only = 0.0; high-resource non-English added (French, German, Chinese, Japanese) = 0.3; mid-resource language as primary (Arabic, Ukrainian,

Czech) = 0.6; low-resource language as primary (< 10M speakers) = 0.8; South Asian language as primary target = 1.0.

$$\text{MEI}(s) = \frac{1}{N(s)} \sum_{p \in s} m(p) \quad (3)$$

Language resource tier for each paper is determined during annotation from the primary target language reported by the authors, mapped to resource levels following Joshi et al. (2020); this constitutes a sixth annotation dimension applied only for MEI computation. Inter-annotator agreement on language tier assignment was $\kappa = 1.00$ across the 30-paper IAA sample, as tier assignment is unambiguous given the language name. The ceiling of 1.0 is reserved for South Asian languages Bengali, Hindi, Urdu, Tamil, and Nepali which collectively serve over 600 million school-age learners (UNESCO, 2023) yet have no dedicated educational NLP system in our corpus. The MEI is intentionally normative rather than purely descriptive: South Asian languages receive the highest weight not because they are the least-resourced by Joshi et al. (2020) criteria, but to foreground the specific disparity between their massive learner populations and their complete absence from educational NLP. Other tiers follow Joshi et al. (2020) resource-level classifications.

4 Taxonomy of NLP Education Research

Our corpus of 137 papers organises into four sectors; Figure 1 maps the full taxonomy and Table 1 summarises the three-metric gap profile.

4.1 Assessment & GEC ($n = 53$, 38.7%)

The largest sector spans *automated essay scoring* (AES) and *grammatical error correction* (GEC), united by their shared function of evaluating and improving student language output.

Automated Essay Scoring. LLMs have largely displaced BERT-era regression models as the dominant AES paradigm. Whereas, recent work exploits zero-shot and few-shot LLM scoring (He and Li, 2024; Stahl et al., 2024), with multi-agent debate emerging as a promising direction: dialectical reasoning across multiple LLM agents has been shown to improve holistic scoring consistency (Jang et al., 2025).

Fairness is an active concern in AES (39.1% of AES papers). LLMs can infer demographic

attributes—race, gender, socioeconomic status—from essay text (Kwako and Ormerod, 2024), and student names alone measurably shift automated scores in ways that reflect race and gender hierarchies (Sakunkoo and Sakunkoo, 2025). Cross-prompt generalisation remains a challenge, with recent work addressing zero-shot cross-lingual transfer (He and Li, 2024) and similarity-based scoring generalisation (Bexte et al., 2025).

Finding. AES achieves $\text{VG} = 0.947$ across 19 papers spanning holistic, trait-based, and LLM-based approaches. Only one paper reports any pre/post learning measurement, limited to a single session.

Grammatical Error Correction. GEC has transitioned from sequence-to-sequence and sequence-tagging models toward LLM-based prompting (Davis et al., 2024; Staruch et al., 2025). Multilingual GEC has expanded substantially: the MultiGEC-2025 shared task covers 12 languages (Masciolini et al., 2025), building on dedicated corpora for Czech (Náplava et al., 2022), Ukrainian (Syvokon et al., 2023), and Arabic (Kwon et al., 2023). Architectural innovation continues with a mixture of experts GEC (Qorib et al., 2024), non-autoregressive decoding (Yakovlev et al., 2023), and syntax-enhanced parsing (Zhang et al., 2022).

Finding. GEC achieves $\text{VG} = 1.000$ across all 23 papers: no study in the GEC sub-corpus includes an empirical measure of whether learners’ grammatical accuracy improved as a result of system use.

Other Assessment. The remaining 11 papers address short-answer scoring (Takano and Ichikawa, 2022), CEFR proficiency classification (Jalota et al., 2022; Imperial et al., 2025), AI-generated text detection in educational contexts (Alamleh et al., 2023), and programming assessment (Dascallescu et al., 2025). All 11 achieve $\text{VG} = 1.000$ and $\text{DRI} = 0.000$ —none tested with real learners.

4.2 Feedback & Writing Support ($n = 26$, 19.0%)

This sector supports the writing process through readability assessment, text simplification, and automated feedback generation.

Readability and Simplification. Readability research has expanded multilingually: Naous et al.

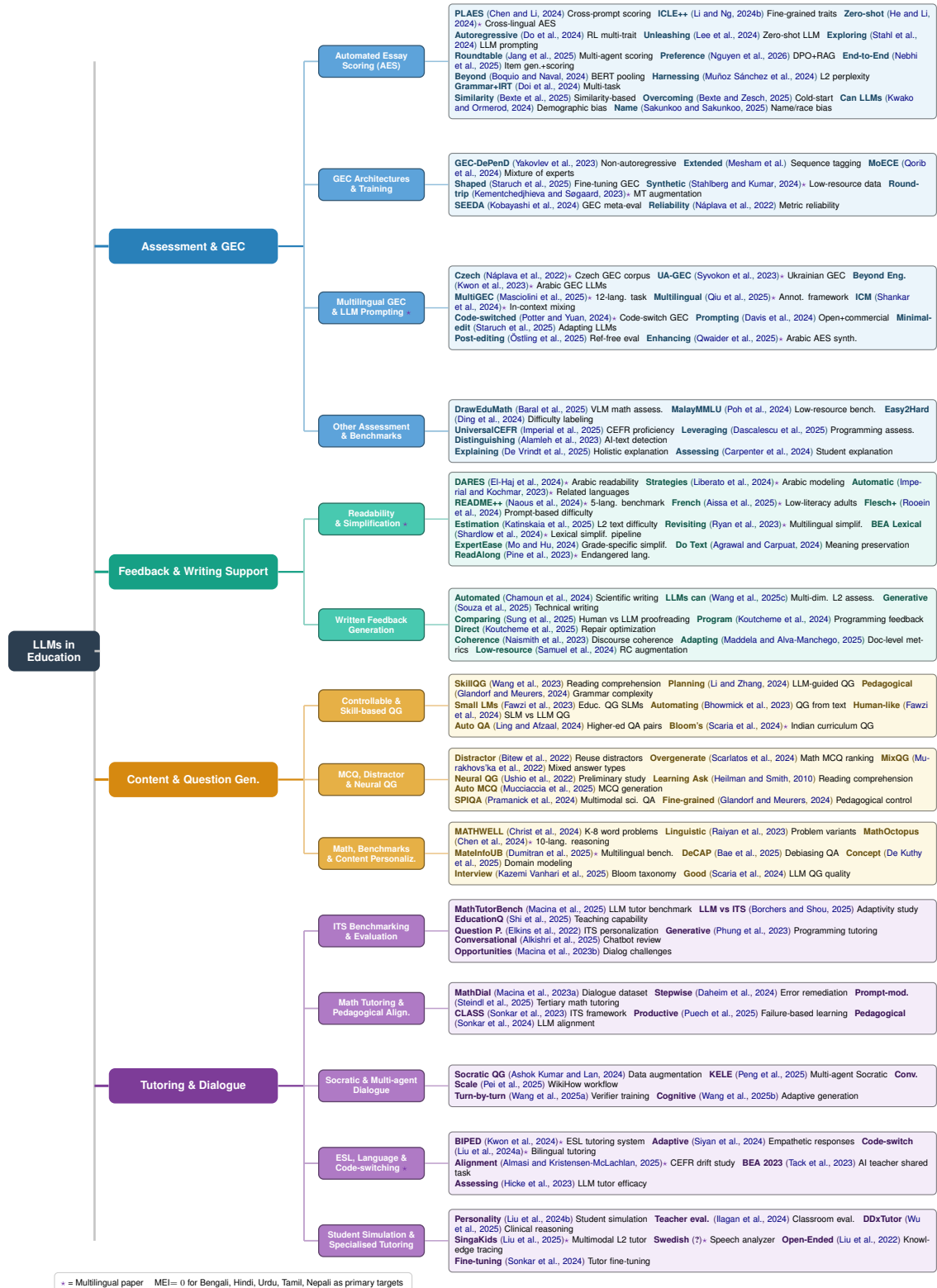


Figure 1: Taxonomy of NLP Education (2022–2025) across four sectors and subcategories.

Table 1: Cross-sectoral gap analysis ($n = 137$, 2022–2025).

Sector / Sub-task	n	%	LLM%	Fair%	Learn%	DRI%	MEI%	VG↑
Assessment & GEC	53	38.7	62.3	39.6	1.9	10.4	12.1	98.1
Automated Essay Scoring	19		57.9	31.6	5.3	17.1	10.5	94.7
Grammatical Error Correction	23		60.9	39.1	0.0	9.8	24.3	100.0
Other Assessment	11		72.7	54.5	0.0	0.0	12.7	100.0
Feedback & Writing Support	26	19.0	69.2	30.8	0.0	16.3	11.9	100.0
Readability & Simplification	13		69.2	38.5	0.0	1.9	37.7	100.0
Writing Feedback & Programming	13		69.2	23.1	0.0	30.8	3.8	100.0
Content & Question Generation	25	18.2	88.0	16.0	4.0	14.0	4.0	96.0
Question Generation (QG/MCQ)	20		85.0	15.0	5.0	16.3	0.0	95.0
Math & Problem Generation	5		100.0	20.0	0.0	5.0	20.0	100.0
Tutoring & Dialogue	33	24.1	90.9	15.2	24.2	25.0	2.7	75.8
Dialogue Tutoring Systems	28		96.4	17.9	21.4	26.8	6.8	78.6
Student Simulation & ITS	5		60.0	0.0	40.0	15.0	0.0	60.0
Field-wide	137	100.0	75.2	27.7	7.3	15.7	12.6	92.7

(2024) benchmark 5 languages on readability estimation, and Liberato et al. (2024) address Arabic school-material readability specifically. Ryan et al. (2023) establish a unified multilingual simplification benchmark, while the BEA 2024 shared task organised a multilingual lexical simplification pipeline (Shardlow et al., 2024). Readability & Simplification achieves the highest sub-sector MEI (0.377) due to this multilingual investment, yet VG = 1.000: no paper measures whether readers comprehend simplified text better.

Writing Feedback and Programming Support.

LLMs are increasingly deployed for feedback on scientific writing (Chamoun et al., 2024), technical documents (Souza et al., 2025), and L2 learner essays (Wang et al., 2025c; Sung et al., 2025). Systems are evaluated on fluency and relevance—never on whether writers improve. Programming feedback work (Koutchme et al., 2024, 2025) addresses automated error explanation and repair suggestion, with evaluation limited to test-set accuracy. **Finding.** VG = 1.000 across both sub-sectors: not one of 26 papers demonstrates that writers improve. Writing Feedback achieves DRI = 0.308—the highest sub-sector DRI—but these evaluate the feedback itself, not the learner’s subsequent performance..

4.3 Content & Question Generation ($n = 25$, 18.2%)

Question generation (QG) dominates this sector. LLM-based QG has largely superseded neural

seq2seq approaches (Murakhovs’ka et al., 2022; Ushio et al., 2022), with pedagogically grounded generation—anchored to Bloom’s taxonomy levels (Scaria et al., 2024; Wang et al., 2023)—representing the most principled research direction. Controllable generation for specific grammar complexity (Glandorf and Meurers, 2024) and spaced-repetition sentence generation (Paddags et al., 2024) represent emerging pedagogical applications.

Math and Problem Generation. Math word problem generation is driven entirely by LLMs (100% adoption): Christ et al. (2024) introduce MATHWELL for K–8 generation, while multilingual reasoning (Chen et al., 2024) addresses language–reasoning intersection. **Finding.** Content & QG achieves MEI = 0.040—the lowest sector MEI—despite generating language-learning materials; every question-generation paper targets English exclusively (MEI = 0.000). VG = 0.960; the single paper reporting a learning outcome measures task accuracy, not skill acquisition.

4.4 Tutoring & Dialogue ($n = 33$, 24.1%)

The most pedagogically mature sector with lowest VG and highest DRI. LLMs have reshaped intelligent tutoring (Borchers and Shou, 2025; Macina et al., 2025), with work on Socratic dialogue (Macina et al., 2023a; Peng et al., 2025), ESL and code-switching (Kwon et al., 2024; Liu et al., 2024a; Siyan et al., 2024), and student simulation

(Tack et al., 2023). Papers measuring learning gains (Liu et al., 2025; Wu et al., 2025; Wang et al., 2025b; Steindl et al., 2025; Puech et al., 2025) cluster here, explaining the lower VG. **Finding.** VG = 0.758, DRI = 0.250—both best-in-field—yet fairness engagement is lowest (15.2%), despite the greatest risk of demographic bias through direct learner interaction.

5 Cross-Sectoral Analysis

Table 1 and Figure 2 present the three-metric gap profile across all sectors.

5.1 The Validity Gap

Field-wide VG = 0.927: 127 of 137 papers never measure whether a real student learned from the proposed system. Only 10 papers report any empirical learning measurement, and 8 of these are in the Tutoring & Dialogue sector. The field has converged on proxy metrics ($F_{0.5}$, Pearson r , BLEU) that measure task performance, not student benefit. DRI correlates with learning outcome at $\rho = 0.425$ ($p < 0.001$), indicating that deployment and validity are structurally coupled.

Representative examples include Liu et al. (2025), who measure vocabulary acquisition gains in elementary students; Steindl et al. (2025), who conduct a three-phase controlled study with university students ($N = 49$); Puech et al. (2025), who validate skill acquisition through a field study with high school students; and Paddags et al. (2024), who report a four-fold increase in word-learning speed for Danish learners ($N = 26$). This clustering suggests deployment pressure—not methodological awareness alone—is the primary driver of learning-outcome measurement, consistent with the DRI–VG correlation ($\rho = 0.425$, $p < 0.001$).

5.2 Deployment Readiness

Field-wide DRI = 0.126. The binary “deployed” count overstates readiness: 77.4% of papers use offline benchmarks only ($d = 0.00$). Year vs. DRI correlates at $\rho = 0.326$ ($p < 0.001$), the strongest temporal correlation in our matrix, driven by 2025 recovery (DRI = 0.231). Assessment & GEC (DRI = 0.075) remains effectively untested with real learners.

5.3 Multilingual Equity

Field-wide MEI = 0.126. Of 137 papers, 101 (73.7%) address English only (MEI = 0.0). Ben-

gali (230M native speakers), Hindi (600M), Urdu (170M), Tamil (80M), and Nepali (16M) have no dedicated educational NLP system in our entire corpus. Content & QG achieves MEI = 0.016—the lowest sector score—despite being the sector most responsible for generating language learning materials.

6 Temporal Analysis

Table 2 and Figure 3 present the temporal gap matrix with paper-level Spearman correlations ($n = 137$).

6.1 The LLM Inflection (2022–2023)

LLM adoption jumps from 45% (2022) to 86% (2023), reflecting ChatGPT (November 2022), GPT-4 (March 2023), and open-weight models. $p < 0.01$. The consequences are immediate: learning outcome measurement drops from 9% to 0%, DRI falls from 0.045 to 0.125 before collapsing to 0.083 in 2024.

6.2 Post-LLM Recovery (2024–2025)

Recovery is visible by 2025: learning rises to 18%, DRI reaches 0.231 (its highest value). However, low-resource language focus rises from 5% (2022) to 9% (2023) then declines to 5% (2025): $\rho = -0.200$, $p = 0.019$. The field is deploying more but serving fewer languages.

6.3 Fairness: Stable but Low

Fairness shows no significant temporal trend ($\rho = 0.054$, $p = 0.530$, n.s.). The field-wide rate of 27.7% reflects a minor but stable concern. Assessment & GEC (39.6%) leads, driven by demographic bias research in AES (Kwako and Ormerod, 2024; Sakunkoo and Sakunkoo, 2025).

7 Research Gaps

7.1 Gap 1: The Validity Crisis (VG = 0.927)

VG = 1.000 in GEC and Feedback means these fields have produced zero evidence of learning benefit. We propose that BEA and AIED reviewers require a *learning plausibility argument*: a brief theoretical account, grounded in learning science, of why the proposed system could improve student outcomes. This is not a demand for a clinical trial—it is a demand for intellectual accountability. Three structural factors explain this gap: NLP publication culture rewards benchmark improvement over classroom impact; the 2023 LLM inflection coincided with a rush to document

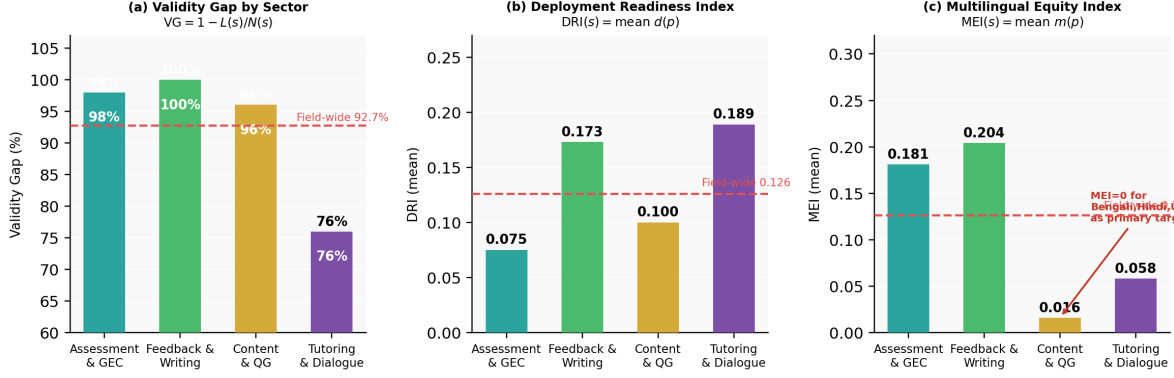


Figure 2: **Three-metric gap profile by sector.** (a) Validity Gap (VG) per sector; dashed line = field-wide 92.7%. (b) Deployment Readiness Index (DRI) per sector; dashed = 0.126. (c) Multilingual Equity Index (MEI) per sector; dashed = 0.126. Colors correspond to sectors throughout the paper. Content & QG achieves the lowest MEI (0.016) despite generating language learning materials. Tutoring & Dialogue achieves the best DRI (0.189) and lowest VG (75.8%).

Dimension	Total (n=137)	2022 (n=22)	2023 (n=22)	2024 (n=54)	2025 (n=39)	Trend	Spearman ($n=137$)
LLM adoption	78.8%	45%	86%	83%	87%	↑↑	$\rho = +0.251^{**}$
Learning outcome (VG)	7.3% (0.927)	9%	0%	2%	18%	≈	$\rho = +0.172^{*}$
DRI (mean)	0.126	0.045	0.125	0.083	0.231	↑	$\rho = +0.326^{***}$
Fairness discussed	27.7%	23%	27%	28%	31%	≈	$\rho = +0.054$ n.s.
MEI (mean)	0.126	0.082	0.136	0.148	0.113	≈	$\rho = +0.047$ n.s.
Low-resource focus	7.3%	5%	9%	11%	5%	↓	$\rho = -0.200^{*}$

Table 2: Temporal gap matrix with paper-level Spearman correlations ($n = 137$). Significance: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. not significant. ✓ >40% ○ 10–40% × <10%. DRI and MEI reported as means (not proportions). Learning collapsed to 0% in 2023 coinciding with peak LLM adoption (86%). DRI has the strongest temporal correlation ($\rho = +0.326^{***}$). Low-resource focus is the only dimension in sustained decline.

capabilities before validating educational benefit (Table 2); and proxy metrics such as $F_{0.5}$ have face validity—correlating with expert quality judgments (Bryant et al., 2019)—but conflate output quality with learning gain (Ferris, 2004; Truscott, 1996). The DRI–VG correlation ($\rho = 0.425$) reflects a structural link: deployment with real learners creates the conditions under which learning-outcome measurement becomes possible, whereas offline benchmarks foreclose it by design.

7.2 Gap 2: Deployment Readiness (DRI = 0.126)

DRI = 0.075 in Assessment & GEC means the largest sector is essentially untested with real learners. Three barriers explain this: *infrastructure* (LLMs require API costs that low-income schools cannot meet), *evaluation complexity* (IRB approval adds months), and incentive structures that currently reward benchmark performance over demonstrated classroom impact. The DRI–VG correlation ($\rho = 0.425$, $p < 0.001$) indicates that addressing deployment is the most effi-

cient path to closing the validity gap.

7.3 Gap 3: Multilingual Equity (MEI = 0.126)

MEI = 0.0 for Bengali, Hindi, Urdu, Tamil, and Nepali represents a critical equity gap: over 600M school-age learners are served by no dedicated educational NLP system in our corpus (Figure 4).

7.4 Gap 4: Cross-Sector Fairness

Fairness (27.7% overall) is critically low in Tutoring & Dialogue (15.2%), the sector with greatest direct learner contact. A chatbot deployed to students without demographic bias analysis is not merely incomplete, but potentially harmful at scale if demographic biases go undetected.

7.5 Gap 5: Learning Science Integration

NLP researchers in education rarely engage with learning science frameworks such as deliberate practice (Ericsson et al., 1993), cognitive load theory (Sweller, 1988), and productive failure (Kapur, 2016), which offer principled predictions about when feedback and tutoring improve learn-

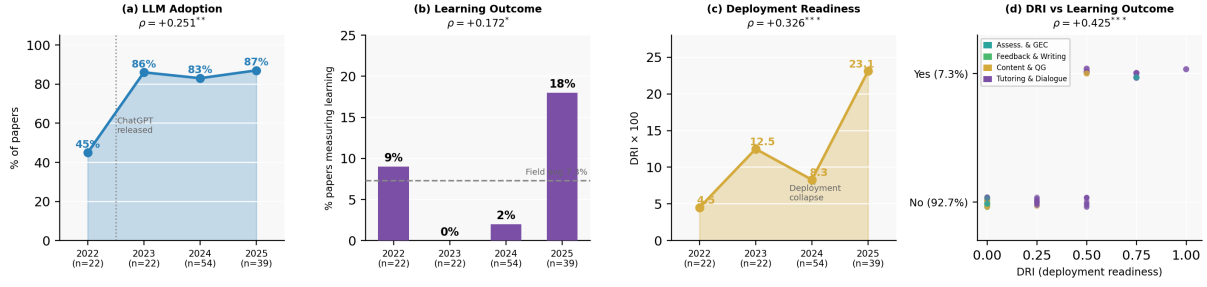


Figure 3: **Temporal trends in educational NLP (2022–2025)**. (a) LLM adoption rose from 45% to 87% ($\rho = +0.251^{**}$). (b) Learning outcome measurement collapsed to 0% in 2023 and recovered to 18% by 2025. (c) DRI shows strongest temporal growth ($\rho = +0.326^{***}$), driven by 2025 deployment recovery. (d) Scatter of paper-level DRI vs learning outcome: $\rho = +0.425$, $p < 0.001$ —the structural link between deployment and learning accountability.

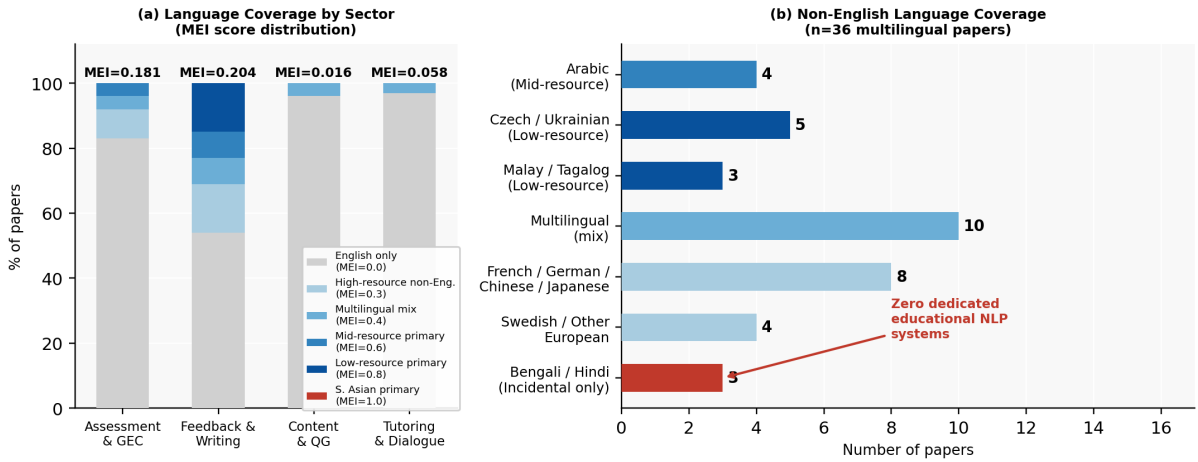


Figure 4: **Multilingual equity gap**. (a) MEI score distribution by sector; Feedback & Writing leads (MEI=0.204) while Content & QG lags (MEI=0.016). (b) Breakdown of the 36 non-English papers by language family. Bengali, Hindi, and Urdu appear only incidentally in multilingual benchmarks—no paper targets any South Asian language as its primary educational medium.

ing. Grounding systems in these frameworks would provide falsifiable hypotheses for validity studies.

8 Conclusion

We present the first quantitative gap audit of educational NLP in the LLM era: 137 papers analysed via Validity Gap (VG=0.927), Deployment Readiness Index (DRI=0.126), and Multilingual Equity Index (MEI=0.126). GEC and Feedback achieve VG = 1.000: zero evidence of learning benefit while South Asian languages score MEI = 0.0 across all 137 papers. Until the field asks “does this system help students learn?”, the validity gap will persist and those who most need effective educational technology will remain unserved.

Limitations

Our search covers NLP-primary venues (ACL Anthology, BEA, AIED, high-citation arXiv); education-primary and HCI venues such as JEDM, IEEE TLT, and CHI are excluded, biasing toward computational contributions and likely undercounting deployed systems and learning-outcome studies. VG and DRI should be read as conservative bounds: the true validity gap may be smaller and deployment readiness higher than reported. The deployment score $d(p)$ reflects the most favourable condition reported. Learning-outcome annotation is binary, collapsing variation in study quality. The MEI is normative: South Asian languages receive the highest weight to surface their population–coverage disparity (Joshi et al., 2020); other weightings may be contested. BERT-era comparisons rely on prior surveys rather than direct annotation.

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